BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 100 · Binary and Denary

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. How many bits are in a nibble and a byte?

Q2. What does an IDE's run-time environment let you do?
A. Repair physical CPU faults
B. Rearrange HDD fragments
C. Manage network user accounts
D. Run and test code

Q3. How do IDE error diagnostics help?

Q4. Two computers receive the same byte but show different characters. Why? What must match?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Binary — Base-two number system using 0 and 1.

Denary — Base-ten number system.

Place value — The numerical weight of a digit's position.

Most significant bit — The leftmost bit with the greatest place value.

Unsigned — A representation with no sign bit.

New learning

Eight-bit place values: Read unsigned place values as 128, 64, 32, 16, 8, 4, 2, 1 from left to right.

Binary to denary: Add the place values in columns containing 1; ignore columns containing 0.

Denary to binary: Start with the largest place value that fits, subtract, then continue until the remainder is zero.

Check your answer: Eight-bit unsigned values range from 0 to 255; verify by adding the selected values.

Worked example

Convert unsigned binary 11001010 into denary. Show the place values used. [4 marks]

1. Write place values

2. The 1 bits are at 128, 64, 8 and 2.

3. Add the selected values: 128 + 64 + 8 + 2.

4. The result is 202 in denary. [4]

We Do · Practise with guidance

Guided 1

Convert unsigned 10110101 to denary. [4]

Worked method: 128 + 32 + 16 + 4 + 1 [2] = 181 [2].

Guided 2

Convert 173 into an eight-bit unsigned binary number. [4]

Worked method: 173 = 128 + 32 + 8 + 4 + 1 [2], so 10101101 [2].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Guided 3

Explain why 256 cannot be represented using eight-bit unsigned binary. [3]

Worked method: Largest eight-bit unsigned value is 11111111 [1], equivalent to 255 [1]; 256 requires nine bits [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the power so the program prints the number of different 8-bit patterns.

```
patterns = 2^...
print(patterns)
```

write

Question 1

Convert the unsigned binary value 01101001 to denary. Show your working. [3]
Source: Original OCR J277-style practice question.

Question 2

Convert denary 154 into an eight-bit unsigned binary number. Show your working. [4]
Source: Original OCR J277-style practice question.

Question 3

State the largest unsigned integer represented by eight bits and explain why. [2]
Source: Original OCR J277-style practice question.

Exit Ticket · Answer before checking

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

An algorithm only gives reliable results when its inputs are handled as specified.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A nibble has 4 bits and a byte has 8 bits.

Q2. D — Run and test code

Q3. They show errors and often their location, helping the programmer find the cause.

Q4. The computers use different character sets or encodings. To show the same letter, both need a matching encoding.

I Do / We Do pages · Guided practice

Q1. 128 + 32 + 16 + 4 + 1 [2] = 181 [2].

Q2. 173 = 128 + 32 + 8 + 4 + 1 [2], so 10101101 [2].

Q3. Largest eight-bit unsigned value is 11111111 [1], equivalent to 255 [1]; 256 requires nine bits [1].

You Do pages · Code First

```
patterns = 2^8
print(patterns)
```

Expected result: 256

You Do pages · Independent questions

Q1. 64 + 32 + 8 + 1 [2] = 105 [1].

Q2. 154 = 128 + 16 + 8 + 2 [2], so 10011010 [2].

Q3. 255 [1], since all eight place-value bits set to 1 total 255 [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: